

PHYS 230 - Electricity and Magnetism

Lecture 3I

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

8 Sources of Magnetic Field

- Magnetic Field of a Current

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

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Vector Magnetic Field of a Current

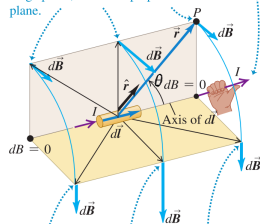
Take an element dq in the current I in figure. Taking dq as a point charge moving with the drift velocity $\mathbf{v}_d$, the magnetic field it generates at a point P a distance r from dq is

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{dq\mathbf{v}_d \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{I \overbrace{d\mathbf{L}}^{d\mathbf{L}} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \mathbf{r}}{r^3}$$

(a) Perspective view

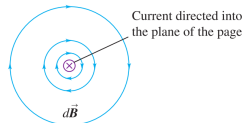
Right-hand rule for the magnetic field due to a current element: Point the thumb of your right hand in the direction of the current. Your fingers now curl around the current element in the direction of the magnetic field lines.

For these field points, $\mathbf{r}$ and $d\mathbf{l}$ both lie in the beige plane, and $d\mathbf{B}$ is perpendicular to this plane.



For these field points, $\mathbf{r}$ and $d\mathbf{l}$ both lie in the gold plane, and $d\mathbf{B}$ is perpendicular to this plane.

(b) View along the axis of the current element



Vector Magnetic Field of a Current

Take an element dq in the current I in figure. Taking dq as a point charge moving with the drift velocity $\mathbf{v}_d$, the magnetic field it generates at a point P a distance r from dq is

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{dq\mathbf{v}_d \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{I \overbrace{d\mathbf{L}}^{d\mathbf{L}} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \mathbf{r}}{r^3}$$

Hence a current I of length $d\mathbf{L}$ (in the direction of the drift velocity $\mathbf{v}_d$) generates a magnetic field $\mathbf{B}$ at a point p a distance r from the current as

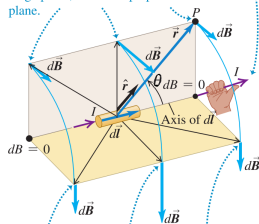
$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \hat{\mathbf{r}}}{r^2}$$

where the vector $\mathbf{r}$ points from the source to the point P : the law of **Biot and Savart**

(a) Perspective view

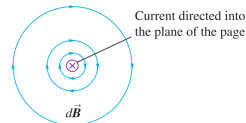
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(b) View along the axis of the current element



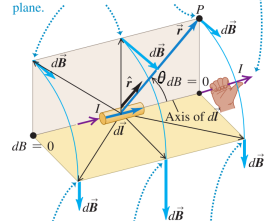
Vector Magnetic Field of a Current

The magnetic field lines of a current are circles centered on the line of current and lying in planes perpendicular to this line

(a) Perspective view

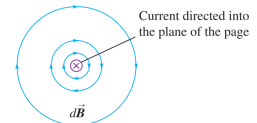
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Vector Magnetic Field of a Current

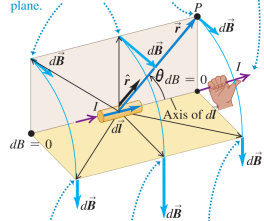
The magnetic field lines of a current are circles centered on the line of current and lying in planes perpendicular to this line

The magnetic field lines are found by the right hand rule: thumb in the “direction” of I and direction of rotation of other four fingers are the direction of the field lines

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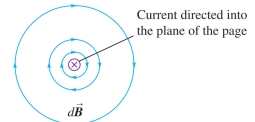
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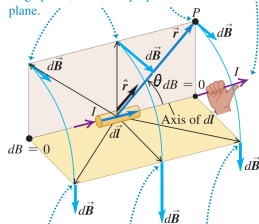
The total magnetic field due to a current is then

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{I d\mathbf{L} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \int \frac{I d\mathbf{L} \times \hat{\mathbf{r}}}{r^2}$$

(a) Perspective view

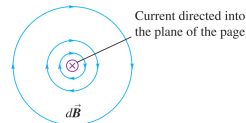
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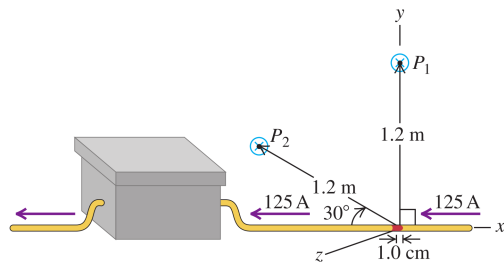
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Exercise I

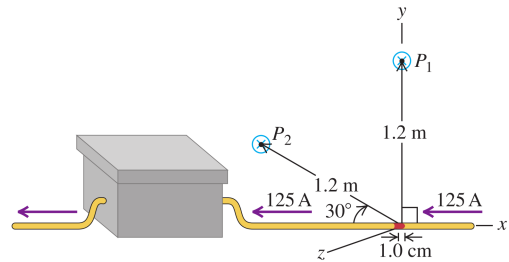
<https://epoll.srv.ualberta.ca> – Code: MBD

A copper wire carries a steady 125 A current to an electroplating tank as in figure. Find the magnetic field due to a 1.0 cm segment of this wire at a point 1.2 m away from it, if the point is (a) point P_1 , straight out to the side of the segment, and (b) point P_2 , in the xy -plane and on a line at 30° to the segment.



Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

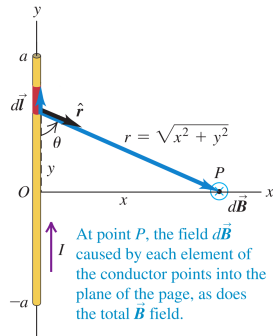


Vector Magnetic Field of a Current

Consider a straight current-carrying conductor of length $2a$. A small element $d\mathbf{L}$ of it (in red) generates a field $d\mathbf{B}$ at a point P a distance x from its center is

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{L} \times \mathbf{r}}{r^3}$$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

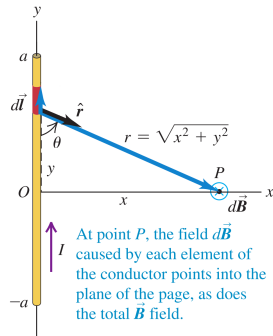


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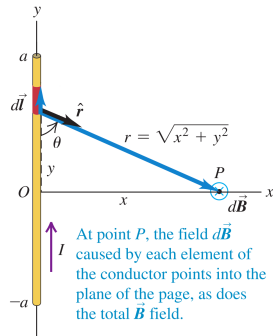


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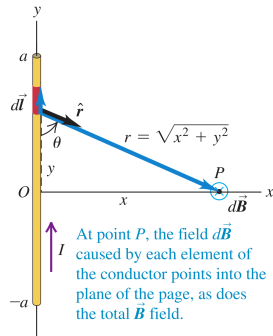


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Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

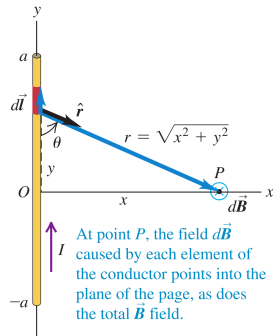


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Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

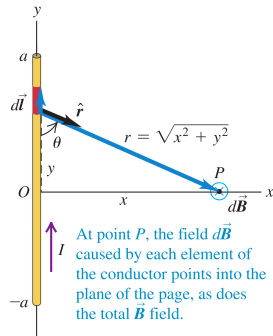


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 &= \frac{\mu_0}{4\pi} \frac{I dy \left[\cos\left(\frac{3\pi}{2} + \theta\right) \hat{\mathbf{j}} \times \hat{\mathbf{i}} + \sin\left(\frac{3\pi}{2} + \theta\right) \hat{\mathbf{j}} \times \hat{\mathbf{j}} \right]}{r^2} \\
 &= \frac{\mu_0}{4\pi} \frac{I dy \left[\cos\left(\frac{3\pi}{2} + \theta\right) (-\hat{\mathbf{k}}) + 0 \right]}{r^2} \\
 &= -\frac{\mu_0}{4\pi} \frac{I dy \cos\left(\frac{3\pi}{2} + \theta\right) \hat{\mathbf{k}}}{r^2} \\
 &= -\frac{\mu_0}{4\pi} \frac{I dy \left[\cancel{\cos\left(\frac{3\pi}{2}\right)}^0 \cos(\theta) - \cancel{\sin\left(\frac{3\pi}{2}\right)}^{-1} \sin(\theta) \right] \hat{\mathbf{k}}}{r^2}
 \end{aligned}$$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

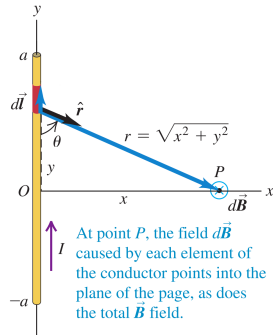


Vector Magnetic Field of a Current

Consider a straight current-carrying conductor of length $2a$. A small element $d\mathbf{L}$ of it (in red) generates a field $d\mathbf{B}$ at a point P a distance x from its center is

$$d\mathbf{B} = -\frac{\mu_0}{4\pi} \frac{I dy \sin(\theta)}{r^2} \hat{\mathbf{k}}$$

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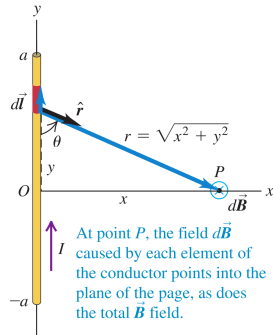


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$$\begin{aligned} d\mathbf{B} &= -\frac{\mu_0}{4\pi} \frac{I dy \sin(\theta) \hat{\mathbf{k}}}{r^2} \\ &= -\frac{\mu_0}{4\pi} \frac{I dy \sin(\theta) \hat{\mathbf{k}}}{x^2 + y^2} \end{aligned}$$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.



Vector Magnetic Field of a Current

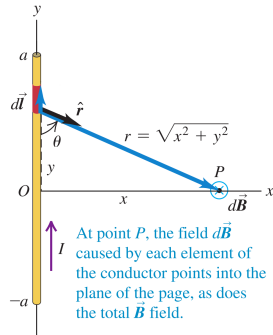
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But we see from the figure that

$$\sin(\theta) = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.



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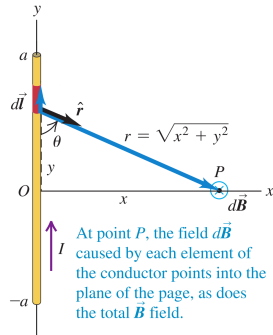
But we see from the figure that

$$\sin(\theta) = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$$

So we can write

$$d\mathbf{B} = -\frac{\mu_0}{4\pi} \frac{I x dy \hat{\mathbf{k}}}{(x^2 + y^2)^{\frac{3}{2}}}$$

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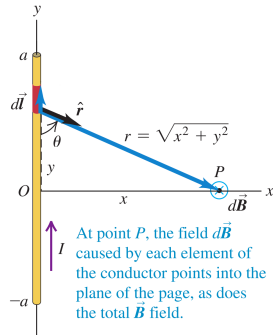


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Integrating this we obtain the total field as

$$\mathbf{B} = \int d\mathbf{B}$$

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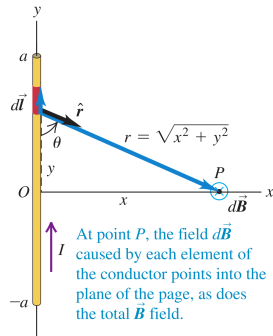


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Integrating this we obtain the total field as

$$\mathbf{B} = \int d\mathbf{B} = -\frac{\mu_0 I}{4\pi} \left(\int_{-a}^a \frac{xdy}{(x^2 + y^2)^{\frac{3}{2}}} \right) \hat{\mathbf{k}}$$

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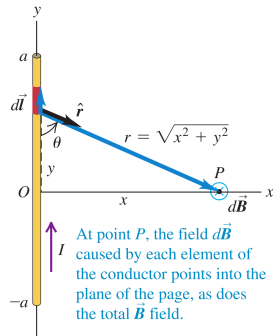


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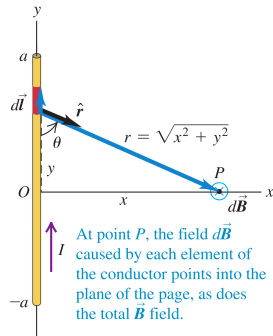


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This field is in the direction of $-\hat{\mathbf{k}}$ only at points P in the xy -plane. The field is actually going around the cable (right hand rule)

Right-hand rule for the magnetic field around a current-carrying wire: Point the thumb of your right hand in the direction of the current. Your fingers now curl around the wire in the direction of the magnetic field lines.

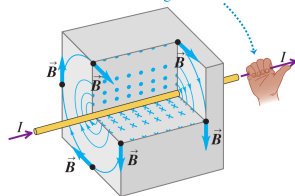
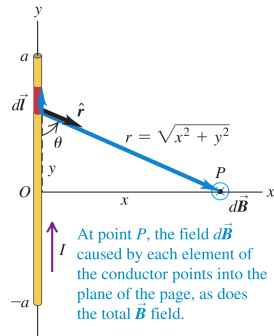


Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

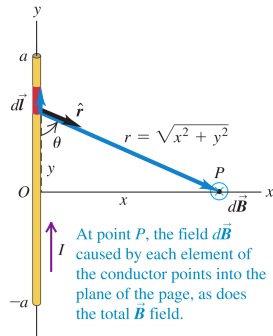


Vector Magnetic Field of a Current

For a very long wire, $a \rightarrow \infty$ and we get

$$\mathbf{B} = \lim_{a \rightarrow \infty} \left(-\frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}} \hat{\mathbf{k}} \right)$$

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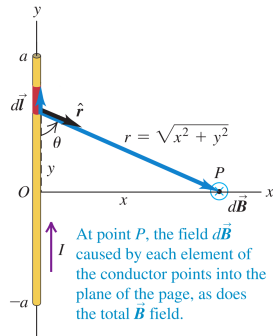


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$$\mathbf{B} = \lim_{a \rightarrow \infty} \left(-\frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}} \hat{\mathbf{k}} \right) = -\frac{\mu_0 I}{4\pi} \lim_{a \rightarrow \infty} \frac{2\phi}{x\phi\sqrt{\frac{x^2}{a^2} + 1}} \hat{\mathbf{k}}$$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

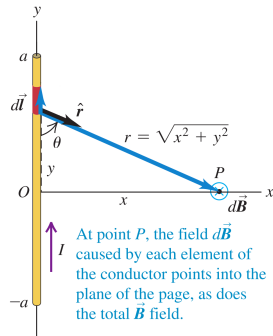


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Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.

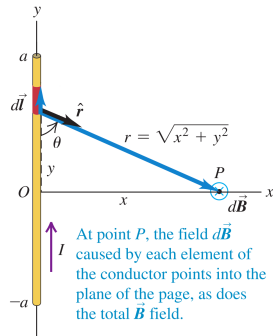


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Vector Magnetic Field of a Current

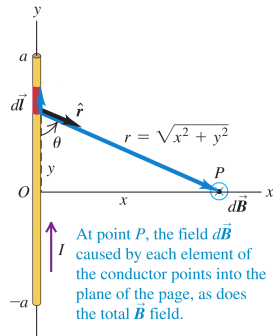
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Again this field goes around the cable. So the magnitude of the magnetic field near a long, straight, current-carrying conductor at a distance x from it is

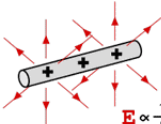
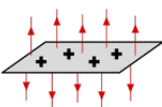
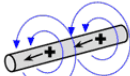
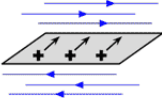
$$B = \frac{\mu_0 I}{2\pi x}$$

Figure 28.5 Magnetic field produced by a straight current-carrying conductor of length $2a$.



Reminder: Electric vs. Magnetic Field Lines

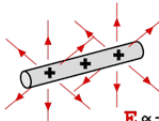
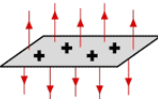
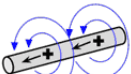
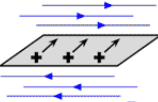
Electric field lines radiate outward from a positive line charge distribution (inward for negative charges). By contrast, magnetic field lines encircle the current that acts as their source.

	point charge	infinite line of charge	infinite plane of charge
electric field $\vec{E}$ units: N/C	 $E \propto \frac{1}{r^2}$	 $E \propto \frac{1}{r}$	 $E \propto 1$
magnetic field $\vec{B}$ units: Tesla (T)	(no magnetic monopoles)	 $B \propto \frac{1}{r}$	 $B \propto 1$

Reminder: Electric vs. Magnetic Field Lines

Electric field lines radiate outward from a positive line charge distribution (inward for negative charges). By contrast, magnetic field lines encircle the current that acts as their source.

Electric field lines due to charges begin and end at those charges, but magnetic field lines always form closed loops and never have endpoints, irrespective of the shape of the current-carrying conductor that sets up the field.

	point charge	infinite line of charge	infinite plane of charge
electric field $\vec{E}$ units: N/C	 $E \propto \frac{1}{r^2}$	 $E \propto \frac{1}{r}$	 $E \propto 1$
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Exercise 2

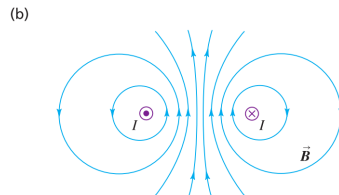
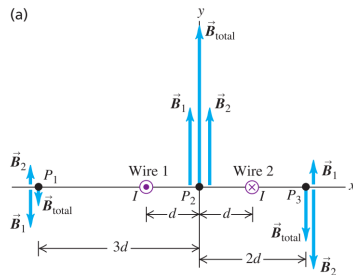
<https://epoll.srv.ualberta.ca> – Code: MBD

A long, straight conductor carries a 1.0 A current. At what distance from the axis of the conductor does the resulting magnetic field have magnitude $B = 5 \times 10^{-5}\text{ T}$ (about that of the Earth's magnetic field on the surface in Edmonton)?

Exercise 3

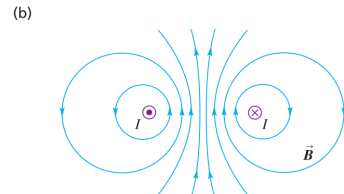
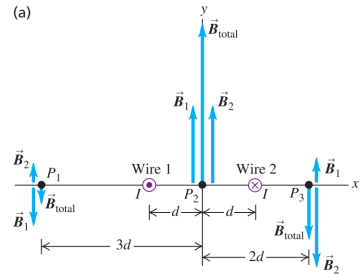
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Figure shows an end-on view of two long, straight, parallel wires perpendicular to the xy -plane, each carrying a current I but in opposite directions. (a) Find $\mathbf{B}$ at points P_1 , P_2 , and P_3 . (b) Find an expression for $\mathbf{B}$ at any point on the x -axis to the right of wire 2.



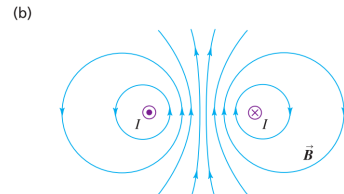
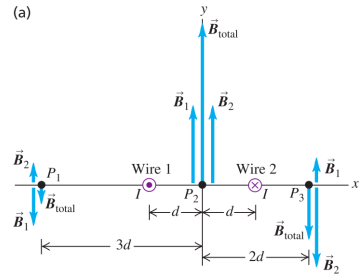
Exercise 3

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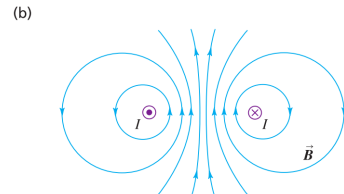
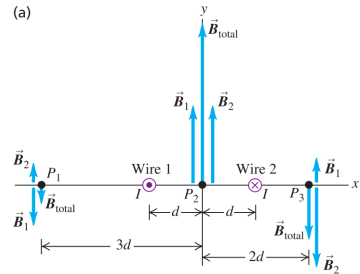
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